CHESSARENA RWANDA
Tournament Pairing, Elo Rating and Fair-Play Monitor
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1. Problem Statement

Organizers of small chess tournaments in Rwanda, including clubs, schools, and universities, often pair players and update ratings manually using notebooks or spreadsheets. This can lead to repeated pairings, incorrect results, inconsistent ratings, and disputes between players. Manual processes also make it difficult to maintain centralized tournament records and enforce fair-play rules consistently.

2. Proposed Solution

ChessArena Rwanda is an Oracle database system developed using PL/SQL to automate chess tournament management. It registers players and tournaments, generates Swiss-system pairings for each round, records game results, and automatically recalculates Elo ratings. Business rules are enforced inside the database using procedures and triggers. Any rejected operation, such as an attempt to repeat a pairing or modify a closed round, is recorded in an audit log.

Main Objective:

To build an Oracle PL/SQL database that automates player pairing, result recording, and Elo rating calculations while enforcing fair-play rules.

Specific Objectives:

1. Store players, tournaments, rounds, and games in a normalized Third Normal Form (3NF) database schema.
2. Generate Swiss-system pairings for each round without repeating opponents.
3. Calculate and update Elo ratings automatically after every recorded result.
4. Prevent rule violations, including repeated pairings and editing closed rounds, while recording such attempts in an audit log.
5. Provide a simple interface for organizers to record results and view tournament standings.

3. Data Source

The project will use the Lichess Standard Chess Games dataset available on Hugging Face: https://huggingface.co/datasets/Lichess/standard-chess-games

A sample of several thousand games will be used instead of the full dataset. Player usernames will be replaced with anonymous player IDs to support data privacy. The dataset is licensed under CC0 1.0 (Public Domain), allowing its use under the applicable public-domain terms.

4. Planned Tables

The database will contain seven main tables:

1. **PLAYERS:** Stores registered players and their current Elo ratings.

2. **TOURNAMENTS:** Stores tournament names, dates, number of rounds, and status.
3. **ENTRIES:** Connects registered players to tournaments and tracks their points.
4. **ROUNDS:** Stores tournament rounds and whether each round is open or closed.
5. **GAMES:** Stores player pairings, game results, and match information.
6. **RATING_HISTORY:** Records player ratings before and after each game.
7. **AUDIT_LOG:** Records blocked or changed actions, including the user and timestamp.

5. Planned PL/SQL Features

The system will implement the following PL/SQL components:

- **Function – calc_new_rating:** Calculates the updated Elo rating after each game.
- **Procedure – generate_pairings:** Uses a cursor to pair eligible players with similar tournament scores while avoiding repeated opponents.
- **Procedure – record_result:** Saves game results and updates player ratings and tournament points.
- **Triggers:** Prevent repeated pairings and prevent result modifications in closed rounds. Rejected attempts are recorded in the audit log.
- **Exception Handling:** Rejects invalid results and operations involving unregistered players.
- **Package – chess_pkg:** Groups the main functions and procedures used by the system.

6. Innovation

The proposed innovation is a user interface built with Streamlit (Python), connected directly to the Oracle database through python-oracledb. The application will call PL/SQL procedures, such as record_result and generate_pairings, and display tournament information and a live leaderboard. All major business rules remain inside PL/SQL, while the application provides an interface for organizers to interact with the system.

7. References (APA 7th Edition)

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